

Name : Syeda Areej E Fatima

Roll no. : 25P0014

Class : BAI-2A

Lab Task : 9

Question 1:

Code:

```
1  #include<iostream>
2  #include<cstring>
3  using namespace std;
4
5  class Person
6  {
7      protected:
8          string name;
9          int age;
10
11     public:
12
13     Person() {}
14
15     Person(int a, string n): age(a), name(n){}
16
17     void set_person_details(string name, int age)
18     {
19         this->name = name;
20         this->age = age;
21     }
22
23     void display_person()
24     {
25         cout<<"Name : "<<name<<endl;
26         cout<<"Age : "<<age<<endl;
27     }
28 };
29
30 class AcademicRecord : public virtual Person
31 {
32     protected:
33         float GPA;
34         string Major;
35
36     public :
37
38     AcademicRecord(float g, string m) : GPA(g), Major(m) {}
39
40     void setAcademicDetails(string n, int a, float gpa , string m)
41     {
42         set_person_details(n, a);
43         GPA = gpa;
44         Major = m;
45     }
46 }
```

```

46
47     void DisplayAcademicDetails()
48     {
49         display_person();
50         cout<<"GPA : "<<GPA<<endl;
51         cout<<"Major : "<<Major<<endl;
52     }
53
54 };
55
56 class Scholarship : public virtual Person, public AcademicRecord
57 {
58     private :
59     int scholarshipAmount;
60     public :
61
62     Scholarship(int a, string n, float g, string m, int sm ):Person(a,n),
63
64     void Evaluate_scholarship()
65     {
66         if(GPA >=3.5 && age < 25)
67         {
68             scholarshipAmount = 50000;
69         }
70         else{
71             scholarshipAmount = 0;
72         }
73     }
74
75     void displayScholarshipDetails()
76     {
77         DisplayAcademicDetails();
78         cout<<"scholarship : "<<scholarshipAmount<<endl;
79     }
80
81 };
82
83 int main()
84 {
85     Scholarship s(20, "Areej", 3.0, "CS", 0);
86
87     s.Evaluate_scholarship();
88     s.displayScholarshipDetails();
89
90     return 0;
91 }

```

output:

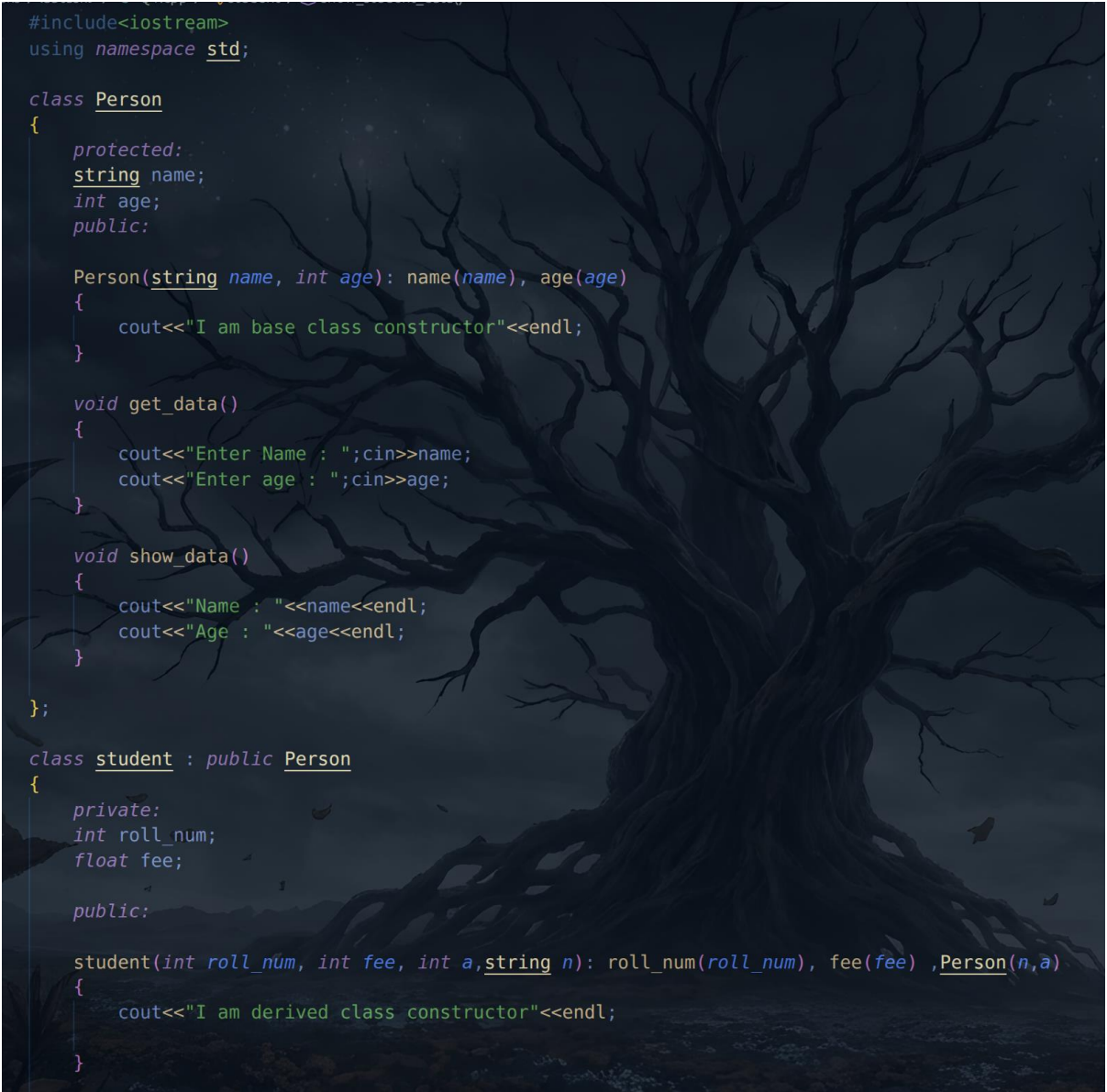
```

Name : Areej
Age : 20
GPA : 3
Major : CS
scholarship : 0

```

Question 2:

Code :



```
#include<iostream>
using namespace std;

class Person
{
    protected:
    string name;
    int age;
    public:

    Person(string name, int age): name(name), age(age)
    {
        cout<<"I am base class constructor"<<endl;
    }

    void get_data()
    {
        cout<<"Enter Name : ";cin>>name;
        cout<<"Enter age : ";cin>>age;
    }

    void show_data()
    {
        cout<<"Name : "<<name<<endl;
        cout<<"Age : "<<age<<endl;
    }
};

class student : public Person
{
    private:
    int roll_num;
    float fee;
    public:

    student(int roll_num, int fee, int a, string n): roll_num(roll_num), fee(fee) ,Person(n,a)
    {
        cout<<"I am derived class constructor"<<endl;
    }
}
```

```

void set_student_data(int r)
{
    get_data();
    roll_num = r;
}

void calculate_fee(float f)
{
    fee = f;
    fee = fee+1000;
}

void show_student_data()
{
    show_data();
    cout<<"Roll number : " <<roll_num<<endl;
    cout<<fee<<endl;
}

};
int main()
{
    student s1( 314, 3000 ,18, "Areej");

    s1.show_student_data();

    return 0;
}

```

Output:

```

lab_tasks/labtask9/" && g++ Q1.cp
I am base class constructor
I am derived class constructor
Name : Areej
Age : 18
Roll number : 314
3000

```